UK CanSat Competition 2026-27

Preliminary Design Review (PDR)

Team Verticality

Lancaster University School of Mathematics

Date: November 2026

1. Team and Mission

1.1 Team Organisation and Roles

There are eight full-time students on the team. We work under our registered Team Leader, who is our Physics teacher. As the rules ask, students are called Student 1 to Student 8. The Student Project Lead runs the plan day to day. The Team Leader stays the official contact for the organisers.

Student	Role	Main jobs
1	Student Project Lead, systems	Plan and schedule, requirement checks, report
2	CAD and mechanical	Shell, internal stack, 3D printing
3	CAD and recovery	Landing legs, parachute sizing and tests
4	Software	Flight code, sensor drivers, logging, ground plots
5	Electronics	PCB design, heater circuit, power budget
6	Radio	LoRa link, packets, frequency change, range tests
7	Ground support and outreach	Ground station, Yagi, outreach plan
8	Biology lead	Algae culture, media, control vials, lab safety

1.2 Missions

Primary mission. After release and during descent, the CanSat measures air temperature and air pressure and sends both to the ground station once every second. We turn the pressure into altitude and plot altitude against time and temperature against altitude. The CanSat has no motor and comes down on a parachute.

Secondary mission: algae as a living shock sensor, with a landing system. Two sealed 1.5 mL vials of the marine alga *Pyrocystis fusiformis* sit in a light-tight bay. The cells give off a flash of blue light when the water around them is shaken or sheared. Chamber A is held at 22 °C by a small heater. Chamber B has no heater, so it sits at the cold ambient temperature of the launch site. Two light sensors log the flashes. A high-g accelerometer logs boost, ejection, parachute opening and landing, so we can match each flash to the event that caused it. Three legs open before touchdown to cut the landing shock.

The science question is simple. Do launch and landing loads make the cells flash, and do warm and cold cells react differently to the same loads? The experiment is new because the cells are both the experiment and the sensor. It also links to real space biology, where living samples have to survive launch.

We only get one flight, so we will not claim statistics from it. To make the result stronger we keep two ground control vials from the same culture, one warm and one cold, that never fly. After the flight we shake them in a controlled way in the lab and compare. We also calibrate the sensors on a vortex shaker before the flight (test T9).

Success criteria.

- **S1** Chamber A stays at 22 ± 1 °C from the pad to landing (from the log).
- **S2** At least 80% of the 1 Hz packets arrive between launch and landing, and the on-board log is complete if the can is recovered.
- **S3** At least one flash above the dark noise is recorded in a chamber. If none is, the control vials must still flash on the ground after the flight, so that "no flash" is a real result.
- **S4** Vials are still sealed after landing (mass change under 5 mg).
- **S5** Legs open before touchdown and the accelerometer records the impact.

2. Software Design

We will write the flight code in MicroPython on the RP2040. It matches the Pico kit and the club code, so the whole team can read it. It runs one main loop with a timer for each job. We do not use the second core, to avoid thread problems. If MicroPython cannot keep up, we move the slow parts to C++ after the CDR.

Modules: sensor drivers (BMP280, ADXL375, NTC, TSL2591), heater PID, radio and packet maker, flight state machine, flash logger, and the ground station receiver and plotter. The accelerometer has its own FIFO, so the code reads it in blocks every 50 ms instead of sampling at 400 Hz by hand. A RAM ring buffer keeps the last five seconds, so the log includes the moments before launch is detected.

Flight states. The code is a simple state machine with five states.

State	What happens	Moves on when
0 Self-test	Check sensor IDs, read the radio frequency from the config file, set the pad pressure P0	All checks pass
1 Pad	Heater on, 1 Hz beacon, ring buffer running. Servo locked out	Altitude above 25 m, or acceleration above 4 g for 100 ms
2 Boost and coast	High-rate logging to flash	Altitude is 5 m below its highest value
3 Descent	Legs release at 150 m above the pad, or 15 s after apogee, whichever is first	Speed under 0.4 m/s for 5 s
4 Landed	Keep sending a beacon and logging for 30 minutes	Switched off by the team

The heater switches off if the battery is below 3.5 V or Chamber A is above 30 °C, so the radio always has power.

Telemetry and frequency. Each packet is a 34-byte binary frame with a CRC (fields in Appendix D). A text CSV line would be about 90 bytes, so binary keeps airtime short. At 433 MHz, spreading factor 7 and 125 kHz bandwidth, one packet takes about 77 ms. That is about 7.7% duty cycle, under the 10% limit we found for this band. Transmit power is set to 10 dBm or less. We will check the exact sub-band against Ofcom IR 2030 before launch. The frequency is stored in a config file and changed over USB before flight (Requirement 14). Changing it by radio on the pad is a stretch goal.

Ground station. A Pico with an RFM96W passes packets to a laptop over USB. A Python script checks the CRC, drops bad packets without stopping, saves two CSV copies and plots altitude, descent rate, temperatures and light counts live. Indicative code is in Appendix A.

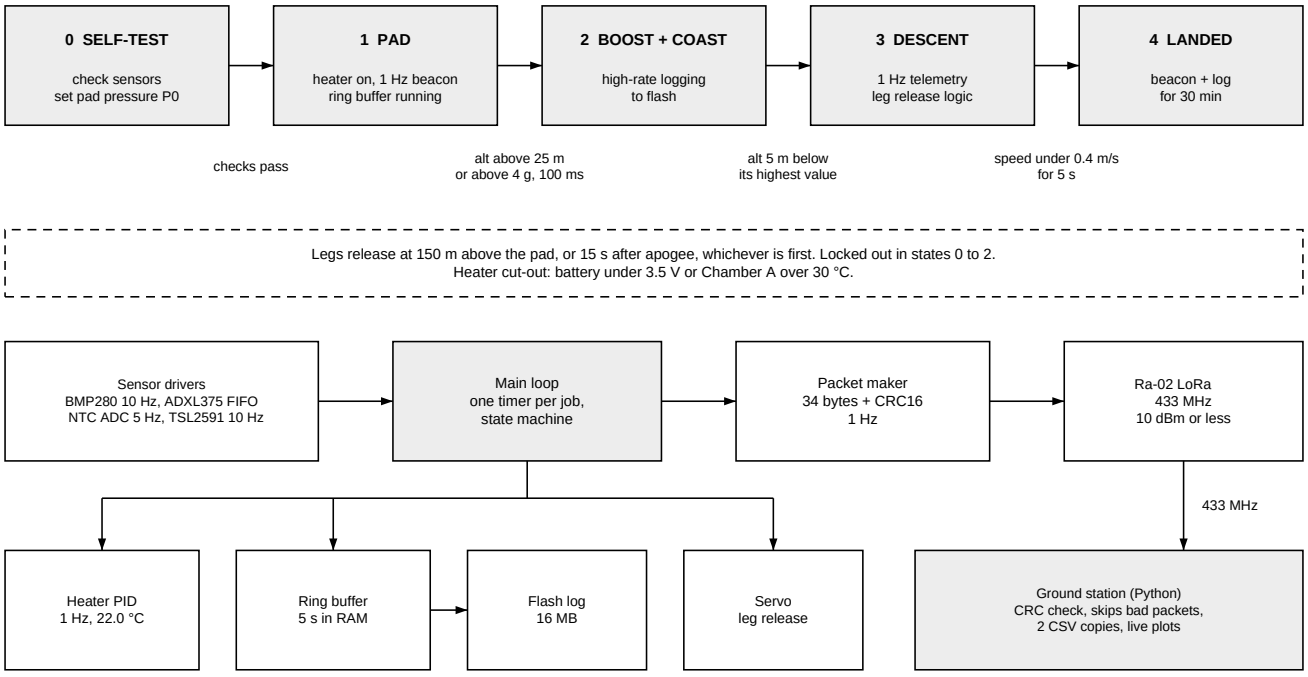


Figure 1. Software structure.

3. Project Planning

3.1 Milestone Plan

Milestone	Target date	Owner	Status
1. PDR submitted	Late Nov 2026	Student 1	In progress
2. Design freeze: shell, schematic, parts list	Early Dec 2026	Students 2, 5	Not started
3. PCB and assembly order placed (Gerber, BOM, CPL)	By mid Dec 2026	Student 5	Not started
4. Version 1 CanSat on Pico and breakout boards, in a printed shell, working	Mid Dec 2026	Students 2, 4	Not started
5. Parachute pull test and drop test done	Dec 2026	Student 3	Not started
6. Algae culture received and on an inverted 12:12 light cycle	Dec 2026 to Jan 2027	Student 8	Not started
7. Custom board arrives, flight code running, 3-hour test passed	Mid Jan 2027	Students 4, 5, 6	Not started
8. CDR submitted with photo of the working CanSat	Late Jan 2027	Student 1	Not started

Nothing is built yet, so only the PDR is in progress. A task-by-task plan with hours will go in the CDR. Regional Launch Events run from late February to March 2027, and ESERO-UK sets our date.

3.2 Team Skills and External Support

	Project mgmt	Electronics	Code	Biology	CAD / print	Radio / outreach
Student 1	●	●			●	
Student 2					●	
Student 3					●	
Student 4			●			
Student 5		●				
Student 6			●			●
Student 7						●
Student 8				●		

● = has the skill now. Student 1 has used KiCad and SolidWorks and has designed fairly complex circuit boards, but not a compact 4-layer one. Gaps we can see: nobody has run a live culture outside a teaching lab and nobody has built a 433 MHz antenna system. Support we have: Lancaster University School of Mathematics (LUSoM) 3D printers that print several materials, science labs with analytical balances, and our Physics teacher. Our school science technician can supervise the culture work, and a school risk assessment for it is planned. We do not have a mentor yet. We will ask for a STEM Ambassador mentor, ideally one with RF or PCB experience, through the STEM Ambassador Programme.

3.3 Top Risks

Risk	Likelihood / impact	How we manage it
Custom board is late or faulty, so we have no working CanSat for the CDR	3 / 4	Build Version 1 on a Pico and breakout boards first, with the same code. Order 5 boards in mid Dec. The board has header pads so a Pico can plug in if the RP2040 circuit needs rework.
The algae do not flash in flight (culture timing, shipping stress, cold, or we cannot get *P. fusiformis* in the UK)	3 / 5	Order cultures early. Keep them on an inverted 12:12 cycle for at least 7 days. Calibrate on a shaker. UK school suppliers sell *Pyrocystis lunula*, which we use as a backup species. Control vials stay on the ground.
Launch provider or ESERO-UK will not clear the live culture, the lithium cell or the legs	3 / 5	Email ESERO-UK early with the cell datasheet, the culture details and a school risk assessment. The cells are non-toxic and sold to schools, but we want it in writing. The mission does not depend on the parachute type, and the legs are locked out until after release.

4. CanSat Design and Initial Build

4.1 Mechanical Design

The shell is a 3D-printed PETG tube, 64 mm outside diameter and 110 mm long (provisional). It fits inside the 66 mm × 115 mm can volume with 1 mm clearance on the diameter and 5 mm on the length. The clear inside is about 100 mm long. That is close to our early 60 × 100 mm layout target, which we keep as a cautious target until the real rocket bay size is known. We will ask ESERO-UK for it. PETG is used instead of PLA because it handles heat and impact better. Printed parts are black PETG with a flexible TPU liner around the vials, which blocks stray light and damps vibration.

Zone (provisional)	Length	What goes in it
Bottom tier	28 mm	Flat lithium cell, servo and release pin, brass ballast, battery hatch and master switch
Middle tier	50 mm	Black bio-chamber block: two vials, heater film, two light sensors, two temperature sensors
Top tier	22 mm	Main board (52 mm round), radio, antenna, vent slots with the BMP280 port

Three tie-rods (M3 stainless) clamp the tiers together. Wires run along the wall and are tied down. The BMP280 faces vent holes in the top bulkhead, because a sensor sealed inside an airtight shell would not measure outside air. The board has slots around the sensor so heat from the board does not reach it. The heater and cell are at least 60 mm below it. We will measure any leftover heating error in test T8.

Legs. Three curved carbon-fibre strips (1.5 mm) sit in shallow grooves along the shell, with ribs inside so the wall stays thick. A nylon loop over a servo pin holds them shut. When the pin is pulled, the springs push the legs out to a footprint of about 140 mm. Nothing sticks out past 66 mm until the can has left the rocket (Requirement 2). The leg shape and spring design are provisional and will be tested.

Mass. Target mass is 325 g (limit is 300 to 350 g, Requirement 3). Our parts estimate adds up to 263 g, so about 62 g of brass ballast goes in the base. That puts the centre of mass about 43 mm above the base, below the middle of the can (55 mm). This helps the can stand upright. With three legs the worst-case tipping angle is about 39°. The full table is in Appendix B. All numbers are estimates until we weigh the parts.

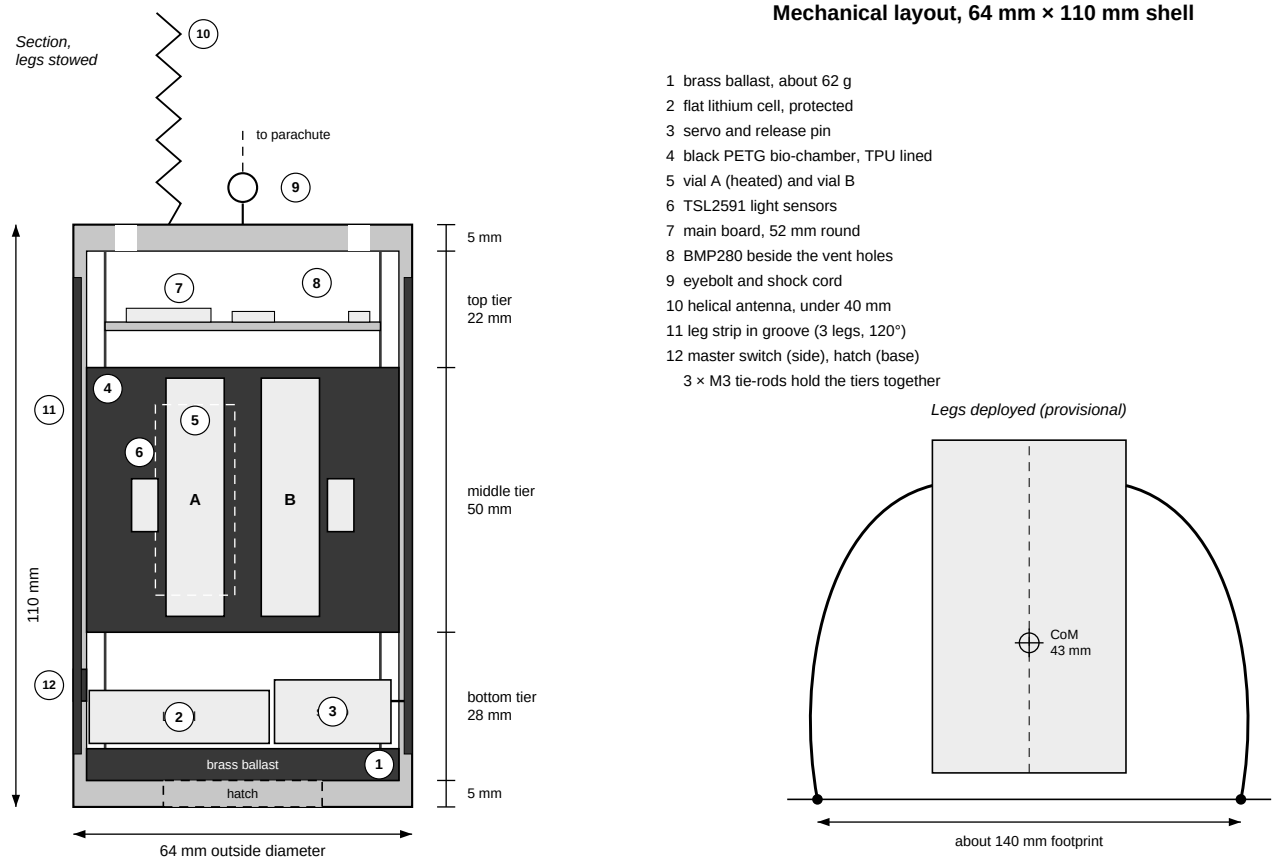


Figure 2. Mechanical layout.

Build so far. Nothing has been built yet. The design is at concept stage, and the numbers in this report come from calculations and research, not from test results. Student 1 will draw the shell in SolidWorks and the board in KiCad in December. Photos and test results will go in the CDR.

4.2 Electrical Design

We will design a round 52 mm, 4-layer board with the bare RP2040, 16 MB flash and a 12 MHz crystal, built by JLCPCB. A printed board means no loose breadboard wires under launch loads. It also saves height compared with a Pico. The board outline is well under JLCPCB's 100 × 100 mm price band. The radio module, servo and cell are soldered or plugged in by us. A Pico can be used in the same place if the board fails.

Sensor	Bus and rate	Why
BMP280 (pressure, air temperature)	SPI, read at 10 Hz, sent at 1 Hz	Primary mission, altitude, leg release
ADXL375 accelerometer (±200 g)	SPI, about 400 Hz from its FIFO	Launch, ejection, canopy and landing loads
2 × TSL2591 light sensor	Two I2C buses (same address), 100 ms integration, 10 Hz at most	Flash counts in Chambers A and B
2 × 10 k NTC thermistor	RP2040 ADC, 5 Hz	Heater control and chamber temperatures
Battery voltage, heater duty	RP2040 ADC and PWM, 1 Hz	Power check and heater energy

The BMP280 is typically good to about 1 hPa and 1 °C [5]. That is about 8 m of absolute altitude error, but changes in pressure are much more accurate, so we measure altitude relative to the pad pressure P0. The temperature reading can be higher than the true air temperature because of heat from nearby parts [5], which is why we test it (T8). The TSL2591 reads all visible and infrared light, not one wavelength. That is fine, because the bay is dark and the only light source is the cells, so we log raw counts, not lux. Its fastest reading time is 100 ms. Algae flashes last about that long, so we measure total light per window, not flash shape. The ADXL375 replaces an earlier ±16 g part, which could not read landing loads.

Power. One 3.7 V flat lithium cell, about 2000 mAh with a protection circuit (roughly 10 × 34 × 50 mm). A 18650 cell is 65 mm long and does not fit our 28 mm battery bay. A 3.3 V low-noise LDO runs the electronics, which keeps switching noise away from the light and pressure sensors. The heater (15 Ω film) is switched by a MOSFET straight from the cell. The servo is switched separately so it draws nothing until release. If the servo is too weak at 3.7 V we add a small 5 V boost. The cell is reached through a bottom hatch, and a master slide switch is on the outside (Requirements 6 and 7).

3-hour budget (Requirement 5). Electronics: about 52 mA (RP2040 25, sensors and flash 12, radio 15), budgeted at 60 mA. Heater: 3.7 V / 15 Ω = 247 mA when on. We assume it is on half the time (124 mA). That is an estimate and test T4 will measure it. Total about 184 mA, so 3 h needs about 550 mAh. Worst case, with the heater on all the time at 4.2 V, is 340 mA, or 1020 mAh. Both are well inside 2000 mAh, even with some loss from the cold.

Radio. Ra-02 at 433 MHz, power set to 10 dBm or less. A quarter-wave whip is about 17 cm long and may not fit the rocket, so we plan a short helical antenna (under 40 mm) on the top face, tested for range. We still need to check the bay size.

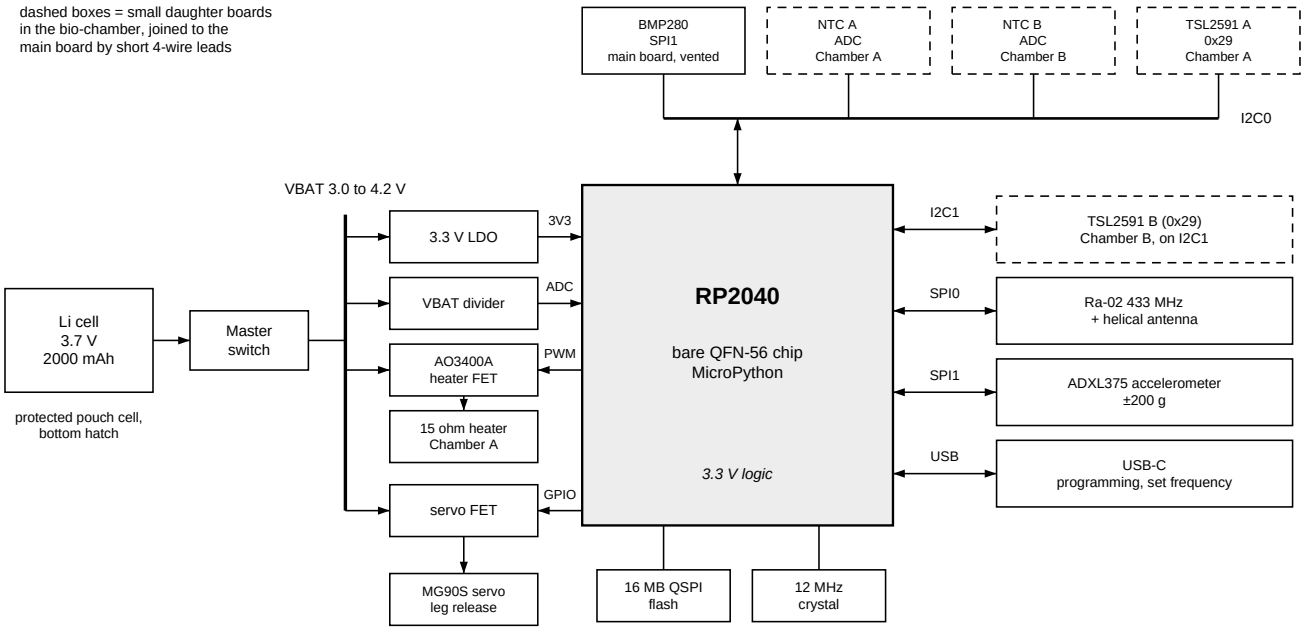


Figure 3. Electronics.

4.3 Landing and Recovery System

Parachute. A bright orange ripstop nylon cross (cruciform) parachute on a 200 N Kevlar shock cord with a ball-bearing swivel. We size it so that drag equals weight at the target speed: $A = 2mg / (\rho \cdot Cd \cdot v^2)$. With m = 0.325 kg, g = 9.81 m/s^2, air density ρ = 1.225 kg/m^3, Cd = 0.75 (an assumption we will measure) and v = 11.5 m/s, A = 6.38 / 121.5 = 0.0525 m^2 (525 cm^2). A cross of five equal squares of side W has $A = 5W^2$, so W = 10.2 cm and the span is about 30.7 cm. 11.5 m/s is inside the 10 to 15 m/s range and above the 10 m/s minimum. The launch provider may swap the canopy. The mission does not depend on the parachute type.

Landing shock. At 11.5 m/s the CanSat has 21.5 J of kinetic energy. If it stops in 15 mm of turf, the mean deceleration is 132.25 / (2 × 0.015) = 4408 m/s^2, about 449 g. The legs add about 55 mm of travel, so the stopping distance is about 70 mm and the mean deceleration falls to about 96 g. That is a best-case mean. The springs also rebound and the peak will be higher. 96 g is still far above the 20 g launch figure, so the cell, vials and board sit in foam or TPU mounts, and we will measure the real peak in test T5. We expect the legs to cut the peak, not remove it. The rounded base and low centre of mass help the can end up upright.

Recovery is not guaranteed (Requirement 12), so the key data goes down the radio link and the can is bright orange.

4.4 Ground Support Equipment

Laptop with the Python dashboard. Ground receiver: Pico plus RFM96W in a printed case, on the same frequency as the CanSat. A hand-held 5-element 433 MHz Yagi for range and for finding the can after landing, and a whip antenna for pad checks. Power bank and a second laptop as a spare. Receiver code is tested with corrupted packets (T7).

Budget (Requirement 13). Our estimate for the flight CanSat is about £275 before VAT, shipping and fees, and about £360 with an allowance for those. The limit is £400. The PCB price uses a live JLCPCB quote (5 four-layer boards, 100 × 100 mm, £5.96). Assembly fees use JLCPCB's published Economic PCBA rates. The full breakdown and assumptions are in Appendix C. Ground equipment is not counted, as the rules allow.

5. Testing

All tests are done before the CDR unless stated. Requirement numbers refer to the 2025-26 guidelines. The 2026-27 registration guide does not list technical requirements, so we use the 2025-26 ones until new ones are issued.

ID	Test	What it checks (Req.)	Pass criteria
T1	Fit and mass: slide the finished CanSat into a 66 mm × 115 mm tube and weigh it	Size and mass (1, 2, 3)	Slides in freely. Mass 300 to 350 g (target 325 ± 3 g). Nothing sticks out with legs stowed.
T2	Parachute pull: hang 5.2 kg (51 N) from the eyebolt, swivel and lines for 5 min	Connection strength (8, 9)	No stretch, tear or damage after the load comes off.
T3	Parachute drop from the college balcony, timed on video over the last 3 m	Descent speed (10)	Speed of 10 to 13 m/s. A canopy needs about 15 m to reach steady speed, so if the balcony is lower the result is a lower limit and we repeat the test at the regional launch.
T4	3-hour endurance in a fridge at about 4 °C for 3.5 h, heater on, 1 Hz packets	Power, heater budget (5, 6, 7)	Packets for over 180 min. Chamber A 22 ± 1 °C. Cell above 3.5 V. We record real heater duty.
T5	Drop at 11.5 m/s (6.7 m), legs stowed and open, dummy mass with water vials, ADXL375 logging	Landing shock (10, 11) and leg claim	Peak is at least 50% lower with legs open. Vial mass change under 5 mg. Electronics still run.
T6	20 g shock: we have no shock rig, so we drop the CanSat onto foam, adjusting the height until the ADXL375 reads 20 g, on each axis	Launch load (11)	No resets, loose parts or broken joints after repeated drops.
T7	Radio range at 400 m line of sight, transmit power 10 dBm or less, with deliberately corrupted packets	Link and ground code (14)	At least 95% of packets arrive. Ground code skips bad packets and carries on.
T8	Air temperature error: BMP280 with heater on and off, compared with a reference thermometer	Primary mission quality	Error under 1 °C, or a known correction.
T9	Dark and flash calibration: shell in bright light for dark counts, then vials on a vortex shaker at about 8 °C and 22 °C	Light sensors and algae	Dark counts stay at the noise floor. Clear flash counts on the shaker. We record the flash threshold.

ID	Test	What it checks (Req.)	Pass criteria
T10	Leg logic: replay a recorded descent into the state machine on the bench	Software	Legs release at 150 m, or at 15 s after apogee, and never on the pad.

For a launch rehearsal we will run the full state machine from switch-on, with a 5 to 10 minute wait on the pad as the guidelines describe, then a tethered drop and a radio range check.

6. Outreach

Our aim is to reach our own school first, then younger pupils nearby, and to share progress online without showing any faces or names. All of this happens before the CDR.

Channel	Who	What	When
School assembly or STEM talk	Students and staff at LUSoM	The mission, the parachute maths and a live demo of the algae glowing when shaken in the dark	Dec 2026
Build log on school channels	School community, parents	Short anonymous updates: CAD, printed parts, tests, the board arriving	Nov 2026 to Jan 2027
Session for local younger pupils	Local primary or Year 9 pupils	How satellites, sensors and biology experiments work, with the algae demo	Jan 2027

Our year cohort is about 150 students, so we aim to reach at least that many in school, plus the younger pupils at the local session. We will count attendance and online views and report them in the CDR.

Use of generative AI and references. Google Gemini and Claude (Anthropic) were used only to check spelling and grammar and to help with the layout and formatting of this report. The mission idea, design, calculations, hardware choices and lab plans are the team's own. Sources: [1] ESERO-UK, UK CanSat Competition Guidelines 2025-2026. [2] JLCPCB, PCB assembly price page, checked Nov 2026, jlcpcb.com/help/article/pcb-assembly-price. [3] Ofcom, IR 2030 licence-exempt short range devices (433.05 to 434.79 MHz). [4] Latz Lab, UC San Diego, "Growing dinoflagellates at home", latzlab.ucsd.edu (culture at 20 to 25 °C, 12 h light and 12 h dark). [5] ESA Education, "Getting started with CanSat: a guide to the primary mission" (T08, 2017).

Appendix A: Indicative flight code

Skeleton only, to show the approach. It is not the final code and has not been run on hardware. Written by the team.

```
# Indicative flight loop (MicroPython on RP2040)
import time

INIT, PAD, BOOST, DESCENT, LANDED = range(5)
state, alt_max, t_apogee, legs_open = INIT, 0.0, None, False

def altitude(p, p0):
    return 44330 * (1 - (p / p0) ** 0.1903)   # barometric formula, metres

def due(name, now, period_ms):
    ...                                        # True once per period

p0 = baro.pressure(); state = PAD              # after self-test
while True:
    now = time.ticks_ms()
    if due("accel", now, 50):                  # read the FIFO in blocks
        for a in accel.read_fifo_g():
            ring.append(now, a)
    if due("baro", now, 100):                  # 10 Hz
        alt = altitude(baro.pressure(), p0)
        alt_max = max(alt_max, alt)
    if due("heater", now, 1000):
        heater.update(chamber_a.temp(), target=22.0, batt_mv=battery.mv())

    if state == PAD and (alt > 25 or ring.peak_g(100) > 4.0):
        state = BOOST; log.dump(ring)          # keep pre-launch data
    elif state == BOOST and alt < alt_max - 5:
        state, t_apogee = DESCENT, now
    elif state == DESCENT and not legs_open:
        if alt < 150 or time.ticks_diff(now, t_apogee) > 15000:
            servo.release(); legs_open = True  # once only
    elif state == DESCENT and speed.under(0.4, 5000):
        state = LANDED

    if due("tx", now, 1000):
        radio.send(pack_packet(state, now, baro, chamber_a, chamber_b,
                               light_a, light_b, ring, battery, heater))
```

Appendix B: Mass and centre of mass (estimates)

Item	Mass (g)	Height from base (mm)
Flat lithium cell with protection	38	14
Brass ballast (sets total to 325 g)	62	8
MG90S servo and pin	14	18
Base, bottom bulkhead, battery hatch	22	10
Bio-chamber block (black PETG, TPU liner)	34	52
Two vials with culture	8	52
Heater film, wiring, sensor boards	9	50
Main board, radio module, antenna	18	92
Shell wall, 64 mm × 110 mm	42	55
Top bulkhead and eyebolt	14	106
Three legs, springs, retention pin	24	50
M3 tie-rods, nuts, screws, switch	20	55
Parachute, shock cord, swivel	20	110
Total	**325**	**43** (centre of mass)

Tipping angle in the worst direction for three legs = atan(35 mm / 43 mm) ≈ 39°, using half the leg footprint radius (70 mm) because three legs form a triangle.

Appendix C: Cost estimate for the flight CanSat

Prices are in pounds before VAT, shipping and fees. Dollar fees from JLCPCB are converted at about £0.75 per $1. Parts prices are estimates and supplier quotes will replace them. JLCPCB fees are from its published Economic PCBA rates, Oct to Nov 2026 [2].

Item	Basis	£
5 × 4-layer boards, 52 mm round (priced as ≤ 100 × 100 mm), 1.6 mm	Live JLCPCB quote	5.96
PCBA setup fee, Economic	$8.18	6.14
Solder paste stencil	$1.53	1.15
SMT joints: about 260 per board × 5	$0.0016 per joint	1.56
Extended-part loading fee: assume 10 unique parts	$3.07 each	23.03
Hand-soldering labour, if any through-hole parts	$3.50 per order	2.63
Parts on the 5 boards (RP2040, flash, BMP280, ADXL375, LDO, crystal, MOSFETs, passives, connectors)	About £15 per board	75.00
2 × Ra-02 433 MHz modules (flight and spare)	About £9 each	18.00
2 × TSL2591 modules and 2 × NTC thermistors	Estimate	30.00
Lithium cell, connectors, master switch, MG90S servo, heater film	Estimate	21.00
Algae culture, nutrient medium, 1.5 mL vials, Parafilm	Estimate, to quote	45.00
Carbon-fibre strip, springs, ballast, M3 hardware, ripstop, Kevlar cord, swivel, eyebolt, helical antenna, filament	Estimate	46.00
Subtotal before VAT, shipping and fees		**275.47**
Allowance for VAT (20%), shipping and fees: about 30%	Assumption	82.64
Estimated total (limit £400)		**358.11**

Not counted (ground equipment): receiver Pico and RFM96W, 5-element Yagi, whip antenna, laptop, about £40 to £45. The estimate depends on the extended-part count and on the ADXL375 price, so we will redo it when the BOM is final. JLCPCB lead time for the quote is 3 to 4 days to build plus 8 to 12 working days by standard shipping, so the order must go in by mid December to have boards by mid January.

Appendix D: Telemetry packet (34 bytes, little-endian, 1 Hz)

Field	Type	Bytes
Packet number	uint16	2
State (0 to 4)	uint8	1
Mission time, ms	uint32	4
Pressure, Pa	uint32	4
Air temperature, °C × 100	int16	2
Chamber A and B temperature, °C × 100	2 × int16	4
Light counts A and B	2 × uint32	8
Peak acceleration, g × 10	uint16	2
RMS jerk, g/s	uint16	2
Battery, mV	uint16	2
Heater duty, %	uint8	1
CRC16	uint16	2
Total		**34**